

BRST Invariance and Chiral Gauge Theories

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We use BRST invariance as a guiding principle in the consistent quantization of gauge theories with Weyl fermions. As in other approaches, the group of gauge transformations is given a dynamical role through a Wess–Zumino action; the choice of the corresponding (quantum) measure determines the number of physical degrees of freedom that it actually represents. © 1989 Academic Press, Inc.

1. INTRODUCTION

The presence of Weyl fermions in a gauge theory is known to produce, at the quantum level, gauge non-invariant interactions [1]. The classically conserved fermion current becomes anomalous and the theory is usually regarded as inconsistent.

In recent years, the possibility of consistent quantization of such anomalous theories has been supported by different investigations. On the one hand, the success of Jackiw and Rajaraman [2] in showing that a consistent, unitary, Lorentz invariant theory emerges from an anomalous two-dimensional gauge theory (the chiral Schwinger model) prompted many investigations of different models for chiral fermions [3]. On the other hand, Faddeev and Shatashvili [4] promoted the idea of modifying canonical quantization of anomalous theories adding new degrees of freedom through a Wess–Zumino action [5] so that the resulting theory would regain gauge invariance.

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Soon after these important developments, it was observed that integration over all gauge field configurations in the Lagrangian path-integral approach makes the Wess–Zumino term naturally arise in chiral gauge models, leading to anomaly free theories [6–8].

Although some results were subsequently obtained following the canonical Hamiltonian formalism [9–11] most of the investigations that followed were developed within the Lagrangian path-integral approach, along the lines of Refs. [6–8]: starting from an integral “over all fields” [12] an integration over the gauge group G factors out in non-anomalous theories and continues to play a dynamical role in anomalous theories, with a Wess–Zumino term as its action.

It is important to note that from this point of view, one has in principle $N = \dim G$ new degrees of freedom while in the original proposal of Faddeev–Shatashvili [4] one supposes that $g(x) \in G$ comprises both “coordinate” and “momentum” variables and so one has $N = \frac{1}{2} \dim G$ physical degrees of freedom. Moreover, the choice of effective action in [4] does not introduce a kinetic term for $g(x)$ (i.e., cohomologically trivial terms are not included in the Wess–Zumino action). Then, the Wess–Zumino term is degenerate and the number of degrees of freedom is still reduced (for example, in $d = 2$, $N = \frac{1}{2}(\dim G - \text{rank } G)$ when no kinetic term is present [10]).

There is another approach to gauge theory quantization based on BRST [13] invariance which has proven very useful as a guiding principle in developing the correct quantization procedure when gauge degrees of freedom [14] are present. Indeed, for supergravity and skew-symmetric gauge theories, where the original Faddeev–Popov procedure presents some difficulties, use of BRST invariance leads to the correct answer in a very elegant and economical way [15].

In this paper we restate the quantization of anomalous gauge theories based on the BRST invariance principle in a general way. As we shall see, relying on BRST invariance alone forces the gauge group G to become dynamical but the choice of a measure over G remains free (in contrast to what happens when following the Faddeev–Popov procedure [6–8]). Then a proposal closer to that advocated by Faddeev–Shatashvili [4] emerges if one decides to integrate over an appropriate Liouville measure.

We regard this result, obtained within the BRST quantization scheme, as a confirmation of the possibility of consistent quantization of anomalous gauge theories already suggested by many results obtained using other approaches [4, 6–8].

2. THE GENERATING FUNCTIONAL

We start from the path-integral version of the generating functional Z ,

$$Z = \int \mathcal{D}A_\mu \mathcal{D}\psi \mathcal{D}\bar{\psi} \mathcal{D}B \mathcal{D}\bar{c} \mathcal{D}c \exp(iS) \quad (1)$$

with

$$S = \int dx (\mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{F}} + \mathcal{L}_{\text{FP} + \text{GF}}) \quad (2)$$

and the Yang–Mills and fermionic Lagrangians given by

$$\mathcal{L}_{\text{YM}} = \frac{1}{2} \text{tr } F_{\mu\nu} F^{\mu\nu} \quad (3)$$

$$\mathcal{L}_{\text{F}} = \bar{\psi}(i\not{D} + \not{A})\psi = \bar{\psi}D_w[A]\psi \quad (4)$$

with $A_\mu = A_\mu^a t^a$ taking values in the Lie algebra of some gauge group G (with generators t^a) and the Weyl fermions in the fundamental representation of G . For definiteness we take ψ to be a left-handed fermion, $\gamma_5\psi = -\psi$. Then $D_w[A]$ can also be written in the form

$$D_w[A] = (i\not{D} + \not{A}) \frac{1 - \gamma_5}{2}. \quad (5)$$

The Faddeev–Popov and gauge-fixing Lagrangian $\mathcal{L}_{\text{GF} + \text{FP}}$ is added “à la BRST,” i.e., as a term which does not change physics since it can be written in the form [14]

$$\mathcal{L}_{\text{GF} + \text{FP}} = -i\delta G \quad (6)$$

with G some functional of the fields and δ the BRST transformation satisfying nilpotency, $\delta^2 = 0$. Ghost fields c , $\bar{c}$ and the auxiliary field B are introduced through the BRST transformation

$$\begin{aligned} \delta A_\mu &= \lambda D_\mu c \\ \delta c &= -\frac{\lambda}{2} (c \wedge c) \\ \delta \bar{c} &= i\lambda B \\ \delta B &= 0 \\ \delta \psi &= i\lambda c \psi. \end{aligned} \quad (7)$$

We have not yet defined the measure in (1). To this end, two important aspects regarding (1) must be considered:

- (i) physically equivalent configurations
- (ii) need of regularization.

In fact, both are concerned with the definition of an appropriate measure in (1).

Concerning the fermionic part of the measure, it can be defined through the relation

$$W[A] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[i \int \mathcal{L}_F dx \right] \equiv \det \tilde{D}_w[A] \quad (8)$$

so that

$$Z = \int \mathcal{D}A_\mu \mathcal{D}B \mathcal{D}\bar{c} \mathcal{D}c \exp \left[i \int (\mathcal{L}_{YM} + \mathcal{L}_{FP+GF}) dx \right] W[A]. \quad (9)$$

In (8) we have written $\tilde{D}_w$ instead of D_w to indicate that a right-handed free fermion has been added in order to have a well-defined eigenvalue problem [16].

3. ANOMALIES

Expression (8) needs a regularization since $\tilde{D}_w$ is an unbounded operator and hence the determinant on the r.h.s. is ill-defined.

Were fermions Dirac fermions, any sensible regularization (e.g., ζ -function method, Heat-Kernel method, etc. [17]) is adequate and automatically gives a gauge-invariant and BRST-invariant answer. On the contrary, for Weyl fermions $W[A]$ is in general gauge variant; under a gauge transformation

$$A_\mu \rightarrow A_\mu^g = g^{-1} A_\mu g + g^{-1} \partial_\mu g \quad (10)$$

it changes according to

$$W[A^g] = \exp(2\pi i F[g, A]) W[A] \quad (11)$$

with $F[g, A]$ a local function of A , called the Wess-Zumino term [5], satisfying the 1-cocycle condition [8]

$$F[g_1 g_2, A] = F[g_1, A] + F[g_2, A^{g_1}]. \quad (12)$$

The infinitesimal version of (11) gives the anomaly $\mathcal{A}$ for the covariant derivative of the fermionic current $\mathcal{J}_\mu^a$:

$$D_\mu \mathcal{J}^\mu = \mathcal{A}[A] \quad (13)$$

$$\mathcal{J}_\mu^a = \bar{\psi} \gamma_\mu t^a \psi$$

$$\mathcal{A}[A] = \lim_{t \rightarrow 0} F[g_t, A] \cdot \frac{1}{t} \quad (14)$$

with $g_t = e^{it\theta}$ generated by an element θ of the Lie algebra of G .

4. THE EFFECTIVE ACTION

We will now define an effective action for A_μ , c , $\bar{c}$, and B in order to determine the remaining integration measure.

Consider the generating functional given by (9). The integration measure and the exponential are trivially BRST invariant while the fermion determinant is not. To see this point, let us define a local coordinatization of gauge potentials A_μ in the form

$$A_\mu = A_\mu(A^*, g). \quad (15)$$

Then, under a BRST change,

$$\delta W[A] = \delta[W[A^*] \exp(2\pi i F[A^*, g])], \quad (16)$$

$$\delta W[A] = W[A^*] \exp(2\pi i \{F[A^*, hg] - F[A^*, g]\}), \quad (17)$$

$$h = 1 + i\lambda c$$

or

$$\delta W[A] \neq 0. \quad (18)$$

However, one has

$$\int \mathcal{D}\mu[g] \delta W[A] = 0 \quad (19)$$

with $\mathcal{D}\mu[g]$ an appropriate invariant measure on the manifold of fields g (see discussion below). Relation (19) can be easily confirmed by performing a change of variables in the first term of the argument in the exponential.

Hence BRST invariance is respected provided an integration over the field g is included in Z at an equal foot with A_μ^* , ψ , c , and B integrations. This integration over the gauge degrees of freedom then takes care of the anomaly, leading to a gauge invariant theory, as can be easily shown.

The effective action to be used is defined through the identity

$$\begin{aligned} \exp(iS_{\text{eff}}[A^*, \bar{c}, c, B]) &= \int \mathcal{D}\mu[g] \exp \left[i \int (\mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{GF}} + \mathcal{L}_{\text{FP}}) dx \right] \\ &\quad \times W[A^*] \exp(2\pi i F[A^*, g]) \end{aligned} \quad (20)$$

and the generating functional reads

$$Z = \int \mathcal{D}A_\mu^* \mathcal{D}\bar{c} \mathcal{D}c \mathcal{D}B \exp(iS_{\text{eff}}). \quad (21)$$

Due to (20), the corresponding BRST charge, defined from the BRST conserved current, is a constant of motion.

We have not yet discussed an important point concerning (19)–(21), namely the choice of an appropriate measure over the fields $g(x)$. In the spirit of Refs. [6–8] where the appearance of $\mathcal{D}\mu[g]$ is related to the Faddeev–Popov procedure for gauge fixing, one is naturally led to the identification [6]

$$\mathcal{D}\mu[g] = \mathcal{D}g \quad (22)$$

with $\mathcal{D}g$ the Haar measure over the group manifold.

In the philosophy of Faddeev–Shatashvili [4] the generating functional (1) should be obtained starting from the Hamiltonian approach and integrating over momentum variables. Then, in adding the $\mathcal{D}\mu[g]$ integration one must consider that one-half of the $\dim G = N$ variables correspond to coordinate variables and one-half to momenta. One then must choose for g an adequate Liouville measure $\Omega(x)$ which can be identified with the closed form [19, 4, 10, 11] related to the Wess–Zumino Lagrangian:

$$\begin{aligned} \mathcal{D}\mu[g] = \Omega_4 = \int d^3x \varepsilon_{ijk} \operatorname{tr}(h^{-1} dh h^{-1} dh h^{-1} \partial_i h h^{-1} \partial_j h h^{-1} \partial_k h \\ - h^{-1} dh h^{-1} \partial_i h h^{-1} dh h^{-1} \partial_j h h^{-1} \partial_k h) \quad \text{for } d=4 \end{aligned} \quad (23)$$

$$\mathcal{D}\mu[g] = \Omega_2 = \int dx \operatorname{tr}(h^{-1} dh \wedge h^{-1} dh h^{-1} \partial_x h) \quad \text{for } d=2. \quad (24)$$

In terms of Ω , the Wess–Zumino action for $A=0$, $F[g, 0]$, takes the form

$$F[g, 0] = \int dt d^{-1}\Omega \quad (25)$$

with d^{-1} representing the antiderivative which relates F with Ω [18].

Apart from the Ω contribution, one can include cohomologically trivial terms (known as normal-parity terms in the anomaly language). These terms are of the form $\mu^2 \operatorname{tr} \int d^4x g^{-1} \partial_\mu g g^{-1} d_\mu g$, $a \operatorname{tr} \int d^2x \partial_\mu g g^{-1} \partial_\mu g$, etc. They automatically give a kinetic energy for the field g . If these terms are ignored, $F[g, 0]$ becomes degenerate (i.e., there is no kinetic term in F). The r.h.s. in (23)–(24) can in principle vanish for $g(x)$ generated by a certain sub-algebra of the Lie algebra of G . In particular, it is easy to see that in $d=2$ [10, 20], $\Omega_2(x)$ given by (24) vanishes for g such that $g^{-1} dg$ is in the Cartan algebra of G , and hence the physical degrees of freedom represented by g are $N = (\dim G - \operatorname{rank} G)/2$.

Although our treatment started from the Lagrangian version of the path-integral generating functional, we hope that the same effective action defined by Eq. (20) can be reached starting from the Hamiltonian version of the path-integral. In that case, $\mathcal{D}\mu[g]$ necessarily represents a Liouville measure for g implying fewer physical degrees of freedom than those arising in the approach of Refs. [6–8]. One

should explore the physical consequences of these choices to decide which is preferable: BRST alone does not determine the particular form of the measure although it requires the inclusion of gauge degrees of freedom in the quantization process. This is the main conclusion of our work.

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